Management Information Systems

Week 4

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Last Week…

- ***Business processes*** are structured sets of activities that organizations use to produce goods or services, and information systems help execute, monitor, and improve these processes.
- IS support ***different management levels*** by providing the right type of information for operational, managerial, and strategic decision-making.
- ***Enterprise applications*** are designed to coordinate multiple functions and business processes.
  - ERP systems integrate the key internal business processes of a firm into a single software system to improve coordination and decision making.
  - SCM systems help the firm manage its relationship with suppliers to optimize the planning, sourcing, manufacturing, and delivery of products and services.
  - CRM systems coordinate the business processes surrounding the firm's customers; KMS enable firms to optimize the creation, sharing, and distribution of knowledge.

Weekly Topics

- **Subject to change without prior notice**

W	Ch	Topic	W	Ch	Topic
1		Orientation Introduction to MIS	9	5	IT infrastructure and emerging technologies
2	1	Information Systems in Global business today	10	6	Foundations of Business Intelligence
3	2	Global E-business and Collaboration	11		*No Class*
4	**3**	**Information Systems, Organizations, and Strategy**	12	11	Managing knowledge
5	9	Achieving operational excellence and customer intimacy	13	12	Enhancing decision making
6	4	Ethical and Social issues in information systems	14	10	Hands-on Session / E-commerce
7	7	Telecommunications, the internet, and wireless technology*	15	13/ 8	Building information systems / Securing information systems*
8		*Midterm Exam*	*16*		*Final Exam*

Chapter 3.

Information Systems, Organizations, and Strategy

Questions to Answer

1. **Which features of organizations do managers *need to know* about to build and use information systems successfully?**

2. **How do *Porter's competitive forces model*, the value chain model, synergies, core competencies, and network economics help companies develop competitive strategies using information systems?**

3. **What is the *impact* of information systems on organizations?**

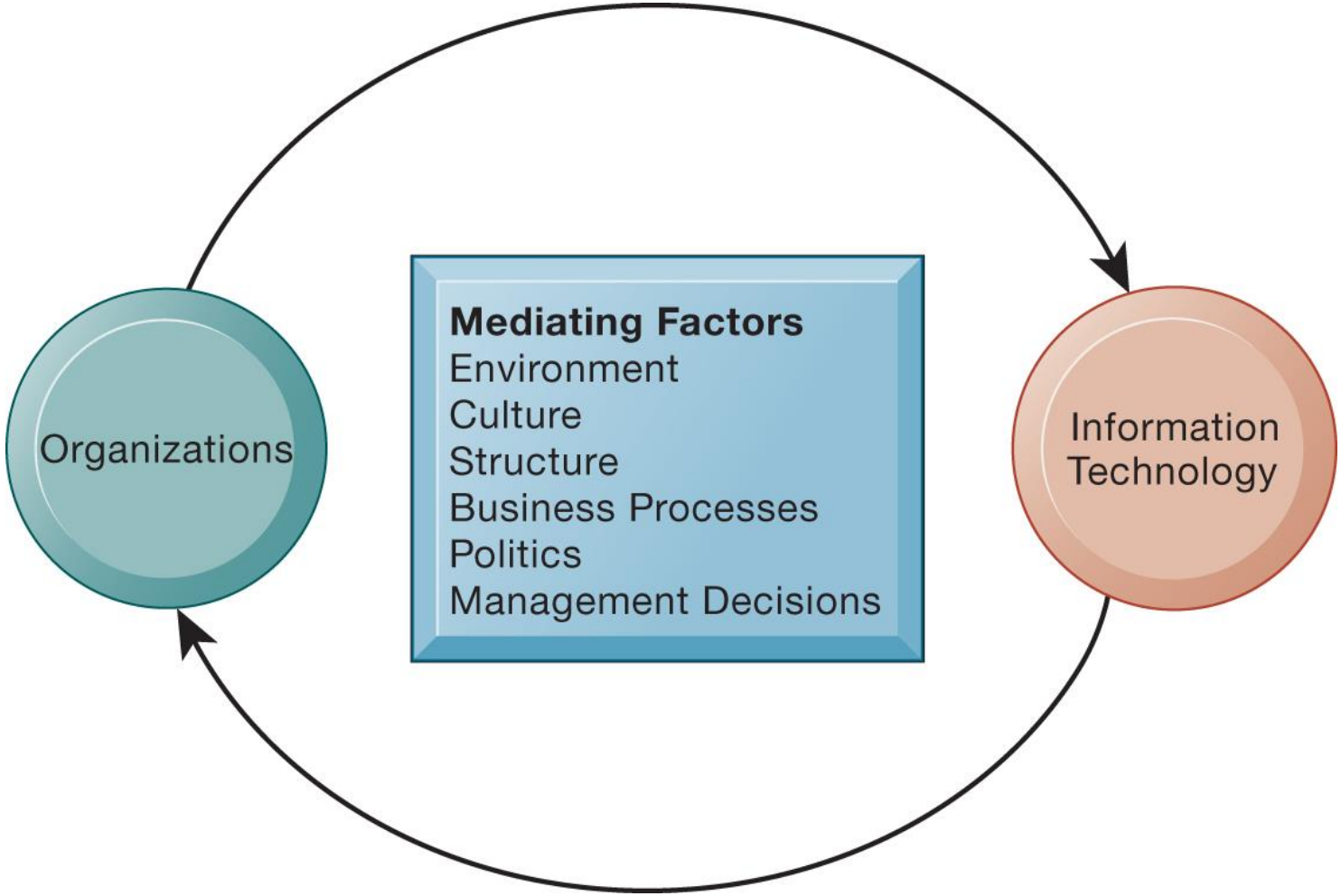
4. **What are the *challenges* posed by strategic information systems, and how should they be addressed?**

3.1

Which features of organizations do managers need to know about to build and use information systems successfully?

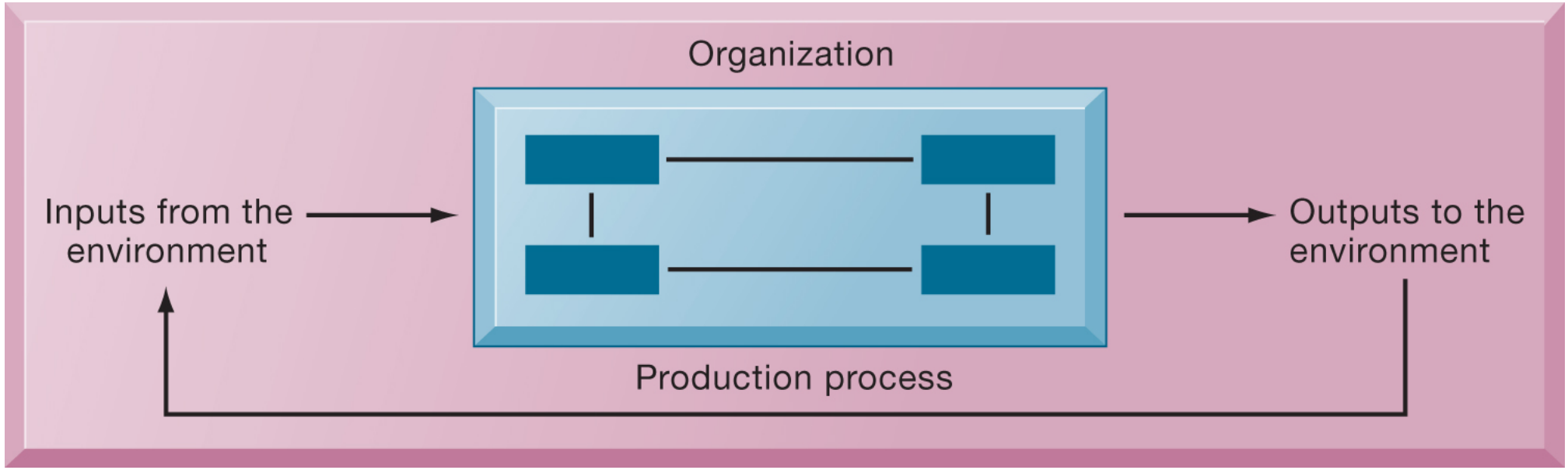
The Relationship Between Organizations and IT

- **IT and organizations influence each other**
  - Relationship influenced by the mediating factors



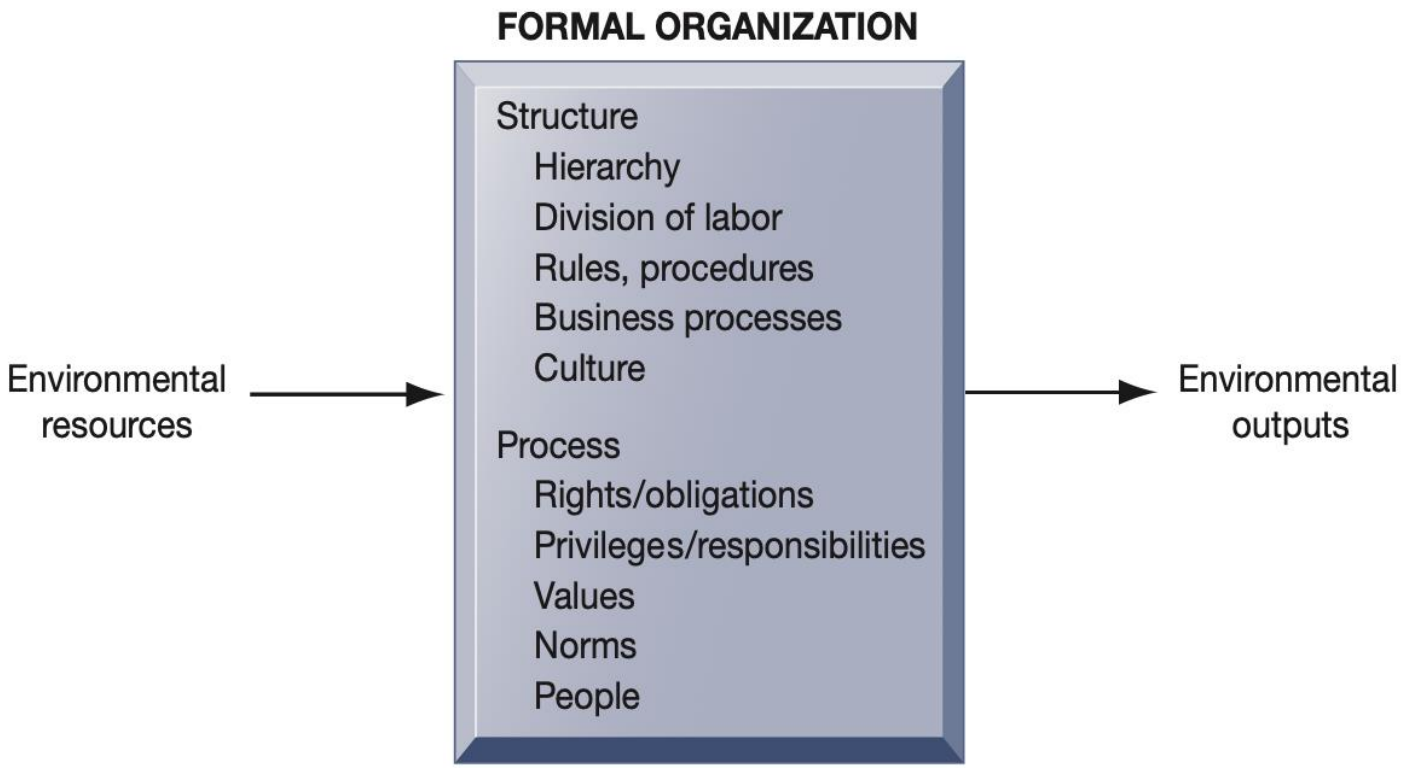
What Is an Organization?: *Technical Definition*

- ***A stable, formal social structure*** **that processes resources from the environment to produce outputs**
  - Stable than an informal group in terms of longevity and routineness
  - A formal legal entity with internal rules and procedures, as well as a social structure



What Is an Organization?: *Behavioral Definition*

- **A collection of *rights, privileges, obligations, and responsibilities* that is delicately balanced over a period through conflict and conflict resolution**



The behavioral view of organizations emphasizes group relationships, values, and structures.

Technical vs. Behavioral Perspectives

- **Technical view:** Focus on the *efficiency* of combining inputs to produce outputs
  - Views firms as adaptable, with capital and labor easily substitutable with technological changes.
- **Behavioral view:** Emphasizes the *impact* of new IS on the organizational balance of rights, privileges, and responsibilities.
  - Acknowledges challenges in changing organizational culture and the need for extensive training and adaptation.
- **Challenges of implementing IS lie in the tension between technological implementation and organizational adaptation**
  - The time lag between technical deployment and effective usage by employees

Features of Organizations

- **Foundational Characteristics**
  - Bureaucratic structure with clear ***divisions of labor*** and specialization
  - Hierarchy of authority with accountability and rule-based decision-making
  - Hiring and promotion based on technical qualifications
- **Principle of Efficiency**
  - Focus on ***maximizing output with limited inputs***
- **Influencing Features**
  - Business processes, organizational culture, politics, etc.

➔ **These features affect the design, implementation, and use of information systems**

Routines and Business Processes

- **All organizations become efficient over time since individuals in the firm develop routines for producing goods and services**
  - Routines are precise rules, procedures, and practices developed to cope with virtually all expected situations
  - Allow the organization to become productive and efficient ➔ cost reduction
- **Business processes are the collections of routines**
- **Business firms are the collection of business processes**

- **New information system applications require that *individual routines and business processes change* to achieve high levels of organizational performance.**

Organizational Politics and Culture

- **People in organizations occupy different positions with different specialties, concerns, and perspectives ➔ Conflicts**
  - ***Political resistance*** greatly hinders organizational change - especially the development of new information systems.

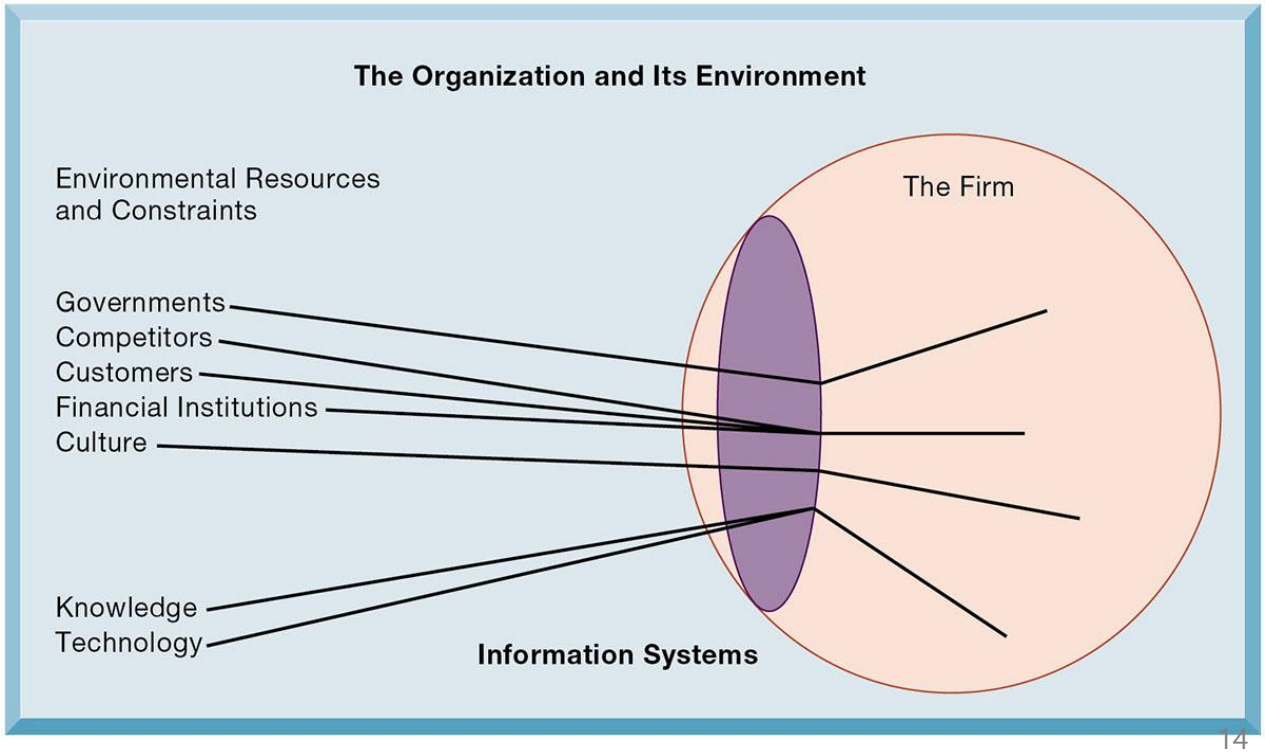
- **Managers who know how to work with the politics of an organization will be more successful than less skilled managers in implementing new information systems.**

- **Organizational culture refers to a set of unwritten rules, and acts as a *unifying force*, reducing political conflict and promotes common understanding and practices**

Organizational Environments

- **Organizations and environments have a *reciprocal* relationship**
  - Organizations are open to, and dependent on, the social and physical environment; Organizations can also influence their environments
- **Environments generally change faster than organizations**
  - Information systems can be an instrument of environmental scanning

"IS as a lens"



Organizational Environments: Disruptive Technologies

- **Definition of Disruptive Technology**
  - Innovations that ***radically change*** the business landscape.
  - Substitute products that outperform existing solutions or extend the market with cost-effective options
- **Examples**
  - Substitution: Car over horse-drawn carriage, word processor over typewriters, iPod over portable CD players, digital over film photography.
  - Market Extension: Small PC hard disk drives offering cheap storage solutions, becoming the dominant market segment.

Image Source: IMDb

Disruptive Technologies' Impact

- **Impact on Businesses**
  - Positive: *"ride the wave"*
  - Negative: those who cannot adapt will disappear
  - Neutral ➔ ?

- **First movers and fast followers**
  - First movers: inventors of disruptive technologies
  - Fast followers: firms with the size and resources to capitalize on that technology

- **Strategic Considerations**
  - Not all innovations are disruptive
  - First movers don't always benefit; "fast followers" may capture greater rewards

Organizational Structure

- **Five basic kinds of organizational structure (Mintzberg)**

ORGANIZATIONAL TYPE	DESCRIPTION	EXAMPLES
Entrepreneurial structure	Young, small firm in a fast-changing environment. It has a simple structure and is managed by an entrepreneur serving as its single chief executive officer.	Small start-up business
Machine bureaucracy	Large bureaucracy existing in a slowly changing environment, producing standard products. It is dominated by a centralized management team and centralized decision making.	Midsize manufacturing firm
Divisionalized bureaucracy	Combination of multiple machine bureaucracies, each producing a different product or service, all topped by one central headquarters.	*Fortune* 500 firms, such as General Motors
Professional bureaucracy	Knowledge-based organization where goods and services depend on the expertise and knowledge of professionals. Dominated by department heads with weak centralized authority.	Law firms, school systems, hospitals
Adhocracy	Task force organization that must respond to rapidly changing environments. Consists of large groups of specialists organized into short-lived multidisciplinary teams and has weak central management.	Consulting firms, such as the Rand Corporation

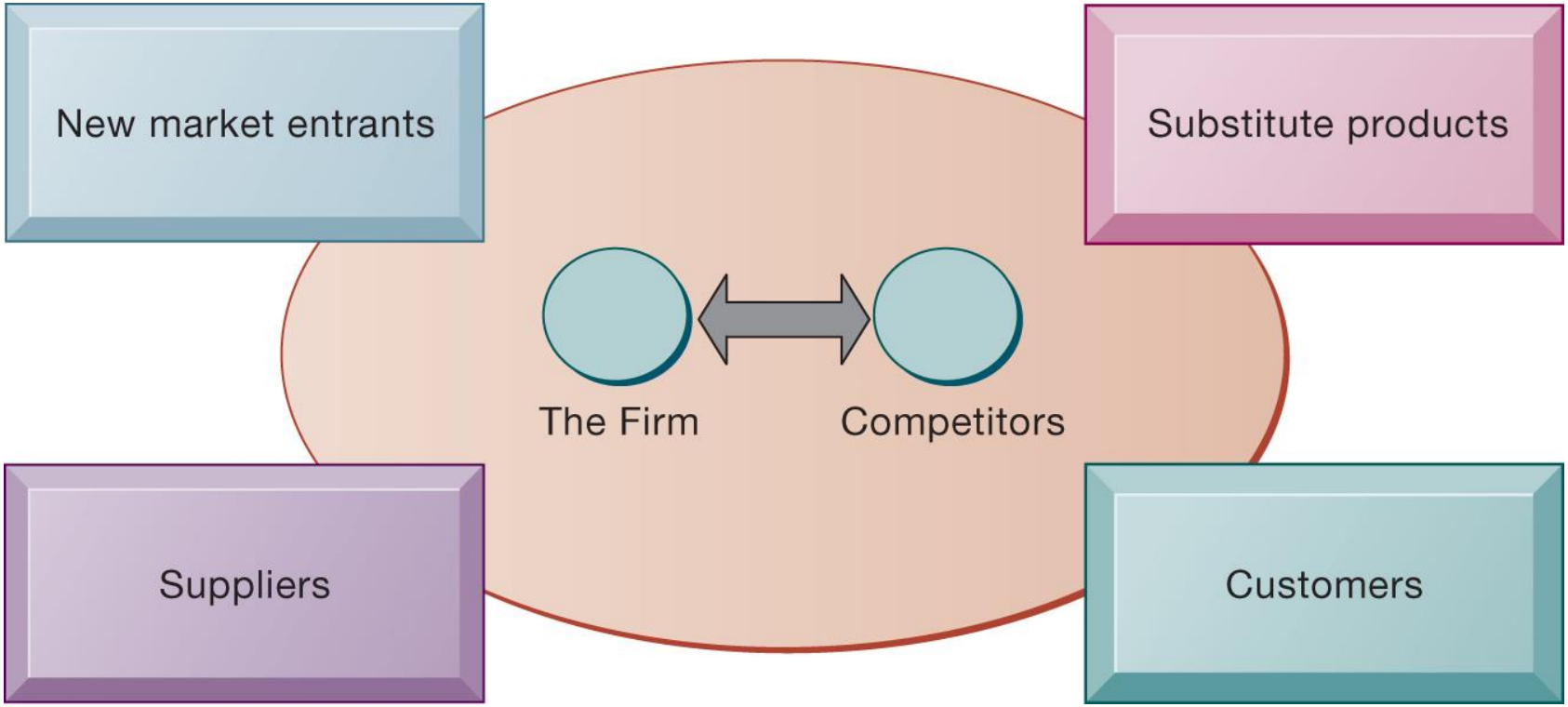
- **Information system often reflects organizational structure**

3.2

How do Porter's competitive forces model, the value chain model, synergies, core competencies, and network economics help companies develop competitive strategies using information systems?

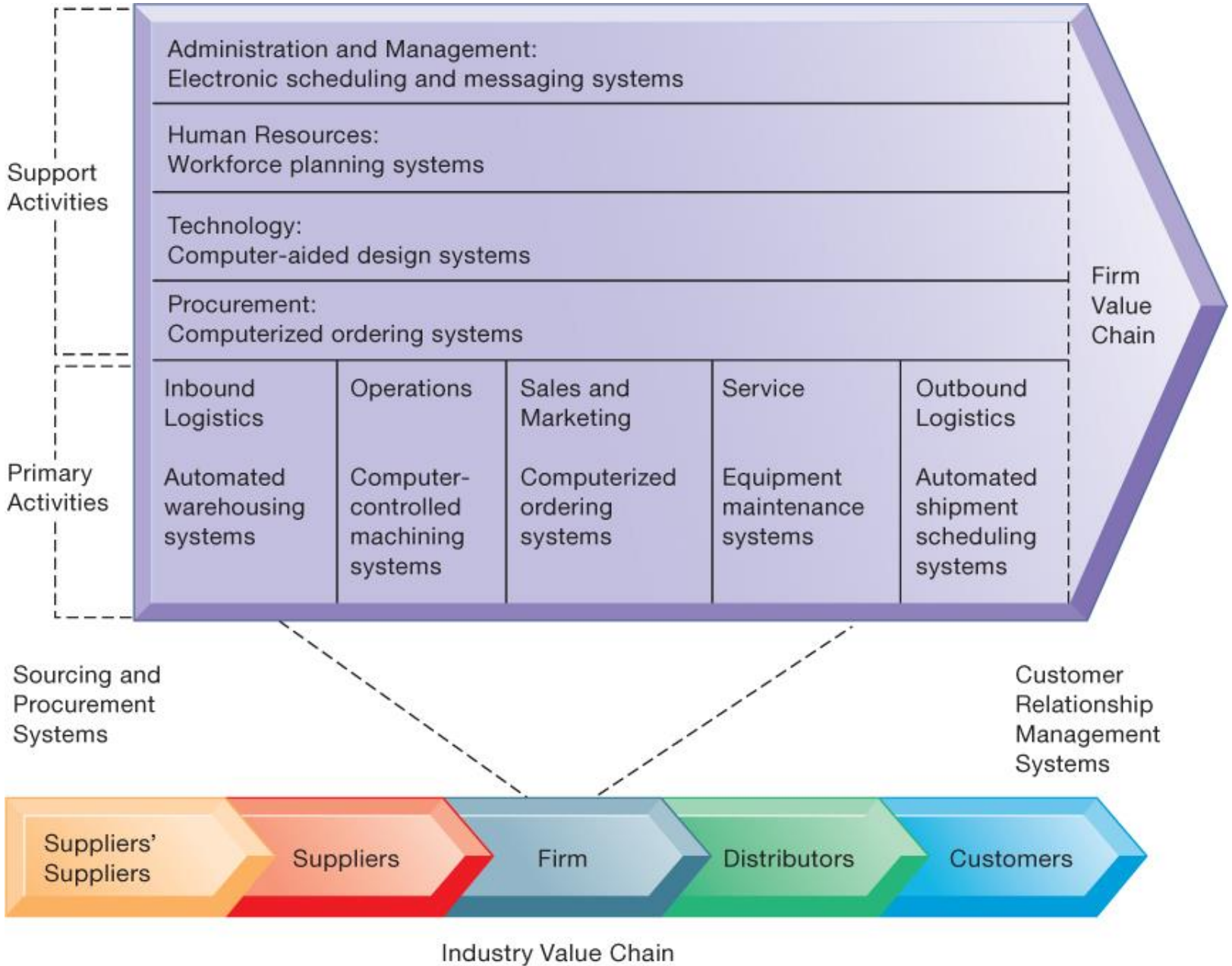
Porter's Competitive Forces Model

- **Provides general view of firm, its competitors, and environment**



- **Traditional competitors**
  - All firms share market space with ***competitors*** who are continuously devising new products, services, efficiencies, and switching costs

The Value Chain Model



The Value Chain Model

- **Helps identify the *key activities* through which a firm creates value and where IT can have the greatest *strategic impact*.**

- **Primary activities are directly involved in producing and delivering products or services.**
- **Support activities enable and enhance the effectiveness of primary activities.**

- **Strategic use of IT focuses on *improving or transforming critical activities* in the value chain to strengthen competitive advantage (e.g., process optimization, cost reduction)**

PORTER'S VALUE CHAIN MODEL: STARBUCKS

Support Activities

Firm Infrastructure

- Company's support system and the functions that allow it to maintain operations
- Includes all legal, administrative and accounting

Human Resources

- Hiring and retaining employees who will fulfill business strategy, as well as help design, market, and sell the product
- Involve in the process of delivering the finished product
- A source of competitive advantage when customer purchasing a service

Technology

- Research and development helps designing the product and improving and automating the process
- For example, a business working towards to reduce the inventory and labor waste by implementing RFID technology in warehouse to improve overall stock accuracy and minimize labor tasks

Procurement

- Process of acquisition of inputs, raw materials or resources, for the firm
- This relates heavily to inbound logistics where company is looking for resale the goods they procure. For example in case of an e-commerce business

Primary Activities

Inbound Logistics

- Starbucks sources high-quality coffee beans from around the world and transports them to its roasting facilities. The company also purchases other raw materials, such as milk and syrups, to be used in its beverages. Focusing on quality allows them to charge higher prices and sell the experience of visiting one of their stores.

Production/Operations

- At its roasting facilities, Starbucks roasts and packages the coffee beans into various blends and flavors. In its stores, the company prepares beverages and food items, trains employees, and maintains equipment and facilities. These centralized facilities allow for economies of scale without sacrificing quality.

Outbound Logistics

- Starbucks distributes its packaged coffee and merchandise products to its retail stores and other distribution channels. The company also uses a network of suppliers and distribution centers to supply its stores with raw materials and other products.

Marketing and Sales

- Starbucks is an excellent example of brand development utilizing marketing efforts, including advertising, social media, and in-store promotions. Their stores are designed to provide a unique and comfortable environment for customers. Employees are trained in providing a particular customer service experience.

Customer Service

- Starbucks places a strong emphasis on providing high-quality customer service, including personalized interactions and customized beverage options. A loyalty program, mobile ordering, and other services enhance the customer experience.

Margin

THE UNITE INNOVATION & TRANSFORMATION MODELS www.digitalleadership.com/UNITE

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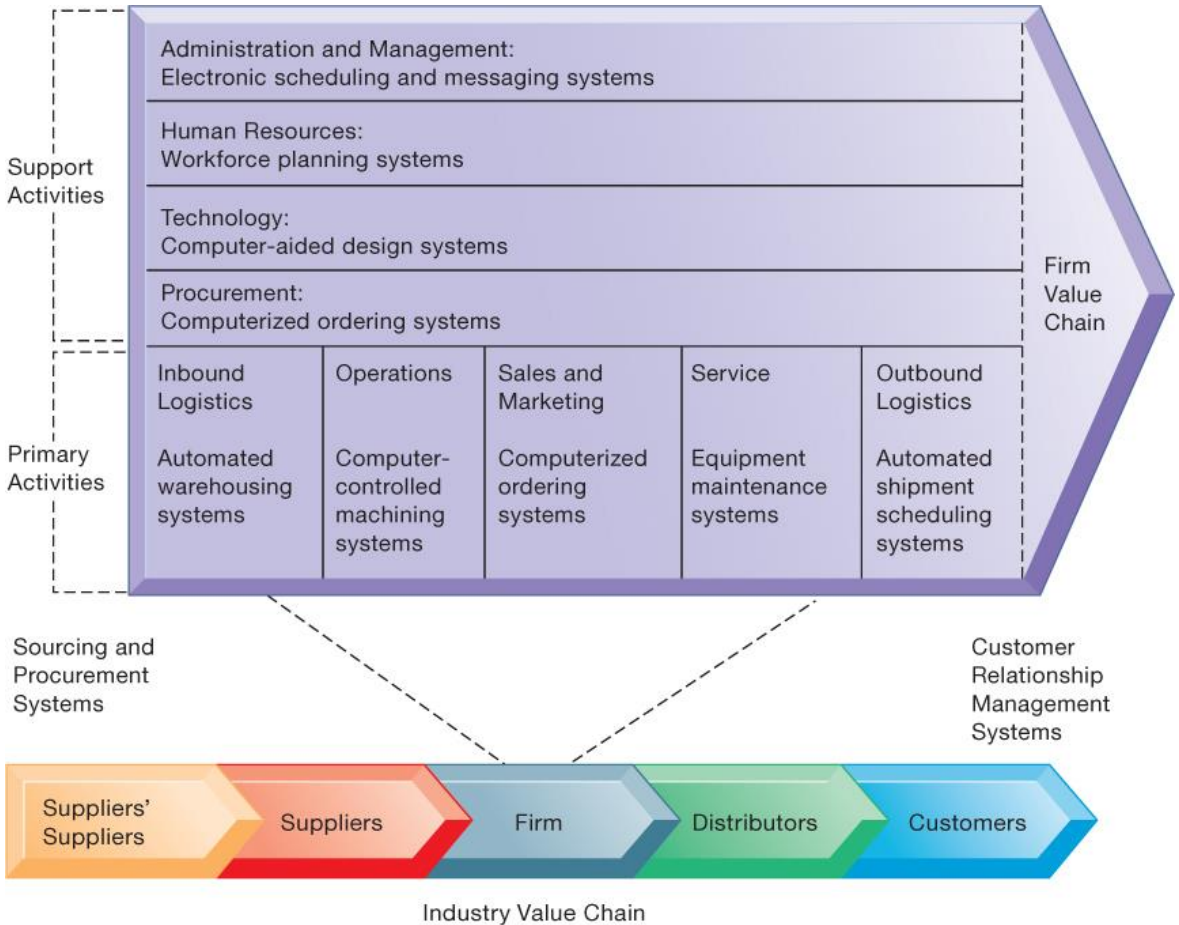
First described by Michael E. Porter in his 1985 best-seller „Competitive Advantage". Designed by: Digital Leadership AG – digitalleadership.com.

Digital Leadership

Source: https://digitalleadership.com/unite-articles/porters-value-chain/

Team Activity #1

- **Use the value chain model to 1) identify and fill in the specific information systems used by a company chosen by your team. 2) Choose one system and provide brief explanation. Submit your work electronically (LMS or email (k.park@postech.ac.kr) or physically.**
  - Split work first, and assign roles.
  - You can use AI to look up info, but don't ask it to draw for you.
  - Be prepared to share yours in next class.



IS Strategies for Dealing with Competitive Forces

- **Four generic *strategies* for dealing with competitive forces, enabled by using IT:**
  1. Low-cost leadership
  2. Product differentiation
  3. Focus on market niche
  4. Strengthen customer and supplier intimacy

STRATEGY	DESCRIPTION	EXAMPLE
1 Low-cost leadership	Use information systems to produce products and services at a lower price than competitors while enhancing quality and level of service	Walmart
2 Product differentiation	Use information systems to differentiate products, and enable new services and products	Uber, Nike, Apple
3 Focus on market niche	Use information systems to enable a focused strategy on a single market niche; specialize	Hilton Hotels, Harrah's
4 Customer and supplier intimacy	Use information systems to develop strong ties and loyalty with customers and suppliers	Toyota Corporation, Amazon

Low-cost Leadership & Product Differentiation

1 2

- **Low-cost leadership**
  - Produce products and services at a lower price than competitors
  - Example: Walmart's efficient customer response system, Alibaba, Temu
- **Product differentiation**
  - Enable new products or services, greatly change customer convenience and experience
  - Example: Google, Nike
  - Mass customization; customer experience management

Amazon: One-click shopping	Amazon holds a patent on one-click shopping that it licenses to other online retailers.
Online music: Apple iTunes	Apple sells music from an online library of more than 50 million songs.
Golf club customization: Ping	Customers can have clubs custom-fit and built to their individual specifications.
Online person-to-person payment: PayPal	PayPal enables the transfer of money between individual bank accounts and between bank accounts and credit card accounts.

Focus on market niche

- **Focus on the specific market segment while utilizing information systems**
- ***Data analysis for marketing***
  - Collecting and analyzing customer data (e.g., buying patterns, preferences).
  - Sources include credit card transactions, demographic data, and online interactions.
- **Example: Hilton Hotels' OnQ system for guest preference analysis and personalized services**
  - CRM systems for intensive data analysis and one-to-one marketing strategies.
  - Creating personalized marketing messages based on individual preferences.
  - Identifying profitable customers for targeted promotions and privileges.

Strengthen customer and supplier intimacy

4

- **Supplier Intimacy (e.g., Toyota and Ford)**
  - Sharing information that enables ***suppliers*** to manage shipments and production, increasing lead times efficiently
- **Customer Intimacy (e.g., Amazon)**
  - ***Personalized*** services that cater to individual customer needs and preferences
- **Increased switching costs make it harder for customers and suppliers to move to competitors**
  - Builds loyalty towards the firm, enhancing long-term relationships

The Internet's Impact on Competitive Advantage

- **Advances in IT have 1) intensified competition, 2) increased customer bargaining power, 3) disrupted many traditional industries, and simultaneously created 4) new markets and successful digital business models.**

Intensification of competitive competition

Increased bargaining power of customers

Industry transformation

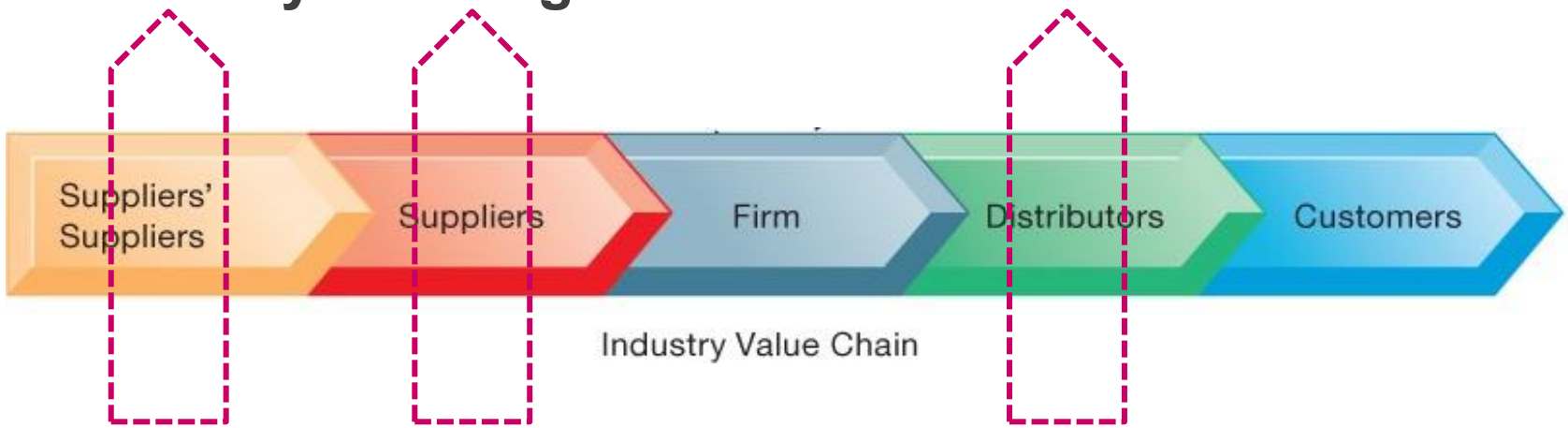
Creation of new markets and opportunities

Using Information Systems for Strategic Advantage

- **Leveraging IS helps strengthen relationships with customers and suppliers by improving coordination and integration.**
- ***Benchmarking*** **compares your business processes against strict standards to evaluate performance.**
- **Industry** ***best practices*** **are proven methods identified by experts and organizations to achieve business goals effectively.**
- **Identifying systems that support value-adding activities and external partnerships helps prioritize applications that offer the greatest competitive advantage.**

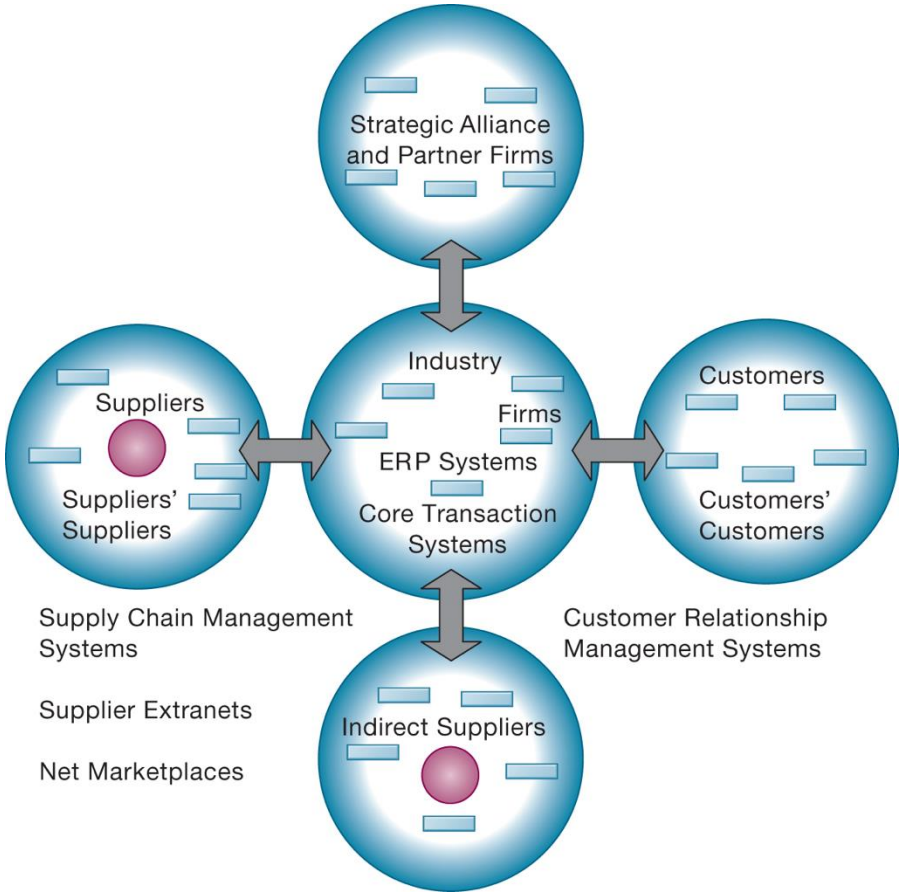
Extending the Value Chain: The Value Web

- Coordination with logistics partners and alignment with customer needs using IS is important
- Developing ***industry standards*** for electronic information exchange increases efficiency, reduces product substitution, and raises entry barriers for new competitors.
- Building industrywide, IT-supported ***networks*** helps firms coordinate activities related to regulation, foreign competition, and industry challenges.



The Value Web

- **Internet enables *highly synchronized industry value chains* called value webs.**
- **Value webs are *networks of independent firms* coordinated through IT.**
  - Allow firms to quickly adjust to changes in supply and demand.
- **Value webs speed up time to market by enabling faster, better coordination among partners.**



Synergies and Enhancing Core Competencies

- **By leveraging IS to consolidate operations, organizations can reduce costs and create a seamless flow where unit outputs become shared inputs**

- **Information systems integrate *merged entities*, providing access to a broader customer base and facilitating *cross-marketing* of products and services**

- **IS as *knowledge enhancers* by enabling effective knowledge sharing and integration across business units**
  - Core competencies are defined as activities where the firm is a leader, gained over many years of practical field experience with technology

Network-Based Strategies

- **Internet and networking technology have inspired strategies that take advantage of firms' abilities to create networks or network with each other.**
  1. Network economics
  2. Virtual company model
  3. Business ecosystems

Network Economics

1

- **Economic value increases as more users join a network, creating *positive network effects* seen in products like telephones, email, and social media**

- **Unlike traditional diminishing returns, network economies exhibit *increasing returns*, with near-zero marginal cost for new users and high marginal gains**
  - Uber, Airbnb, and social networks

- **From a network economy perspective, IS are *strategic assets* that build online communities, strengthen loyalty, and increase value for all participants as the network grows**

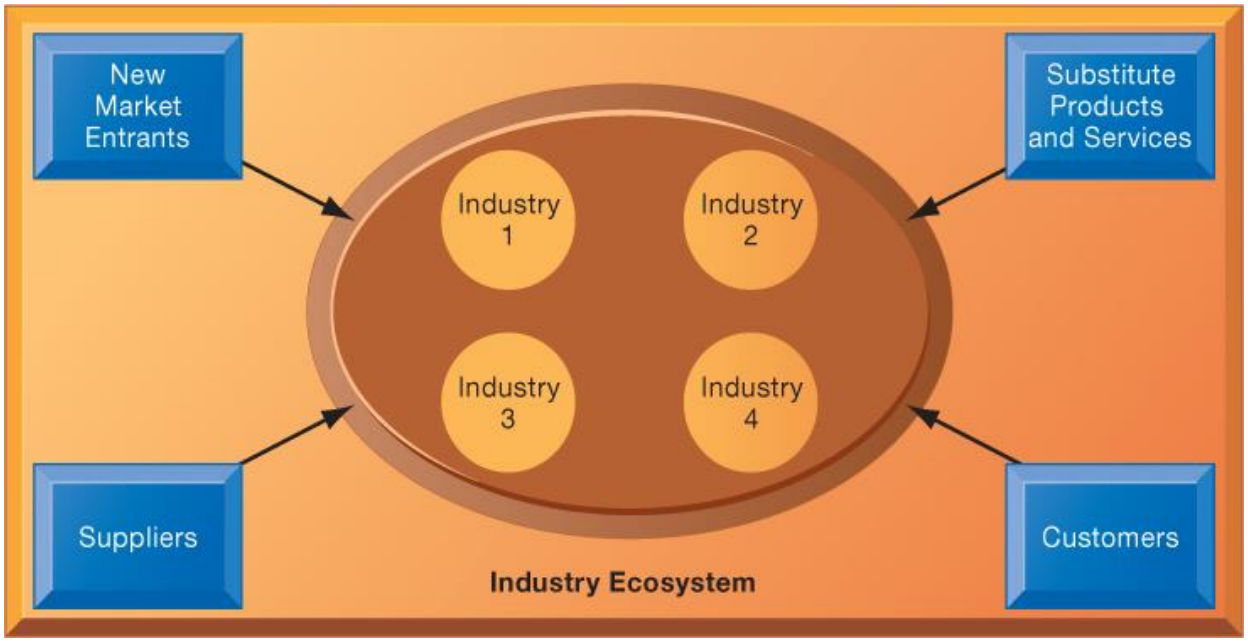
Virtual Company Model

2

- **A virtual company operates as a network-based organization *without traditional boundaries*, connecting people, assets, and ideas through digital networks**
- **The model enables strategic outsourcing and rapid market response by leveraging *external* capabilities instead of building everything in-house**
  - e.g., Li & Fung exemplifies the virtual company by managing global production and shipment through a vast supplier network supported by an extranet
- **Key advantages include high operational flexibility and the ability to design and produce *quickly* in response to changing market trends**

Business Ecosystems and Platforms

- **Business ecosystem is another term for the loosely coupled but interdependent networks of different firms**
  - Suppliers, distributors, outsourcing firms, transportation service firms, and technology manufacturers
- **IT plays an important role in enabling a dense network of interactions among the participating firms.**



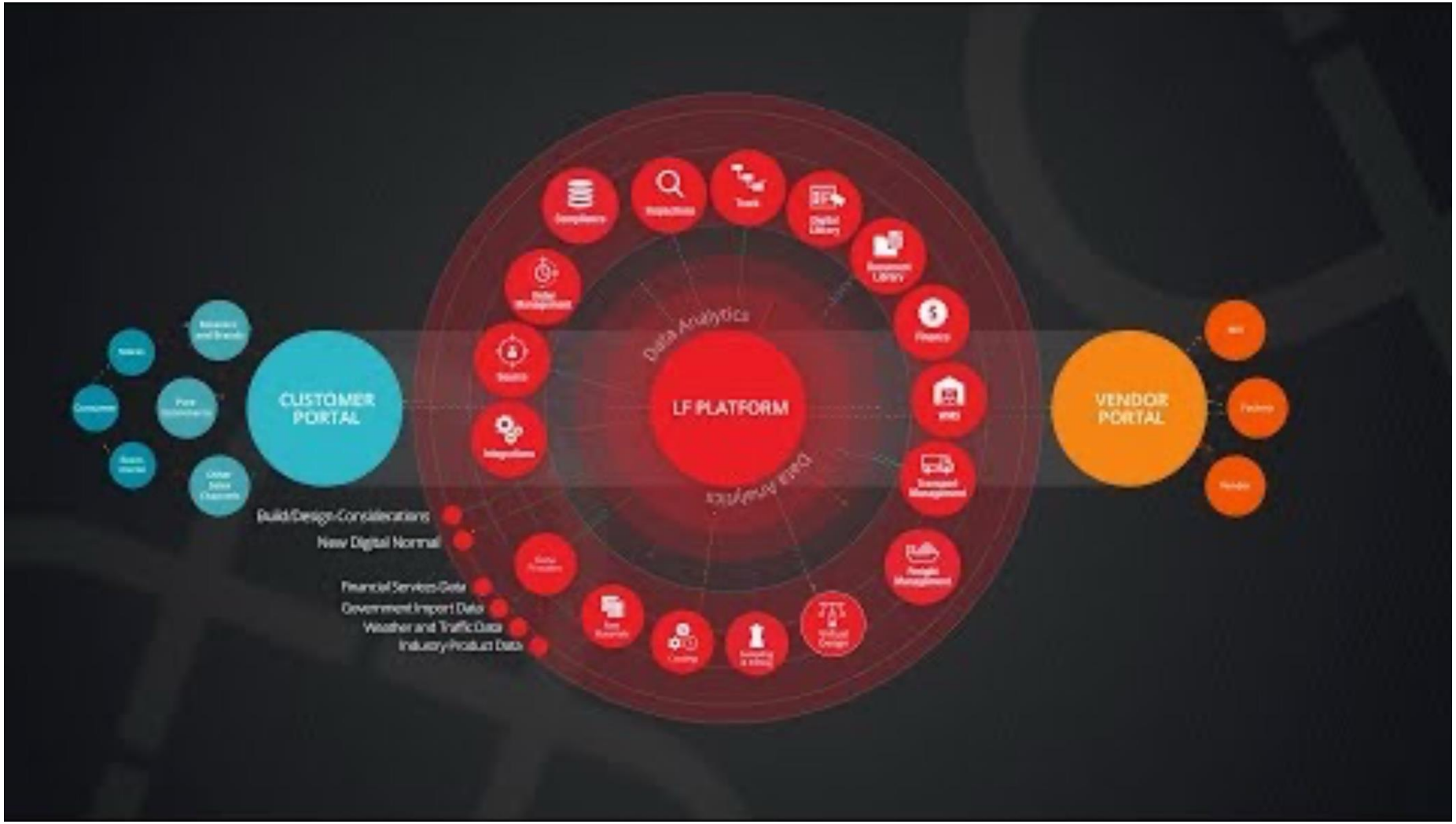
(Iansiti and Levien, 2004).

Business Ecosystems and Platforms (Cont.)

- **An example of a business ecosystem: the mobile Internet platform**
  - Device makers (Apple iPhone, Samsung, LG, and others)
  - Wireless telecommunication firms (AT&T, Verizon, T-Mobile, Sprint, and others)
  - Independent software applications providers
  - Internet service providers
- **One or a few keystone firms in business ecosystems:**
  - ***Keystone firms*** dominate the ecosystem and create the platforms used by other niche firms.
    - Example: Microsoft and Facebook as platform providers enabling other businesses
- **Strategic Implications for Firms**
  - How firms can leverage information systems to become ***keystone players*** or profitable niche contributors.
  - The importance of understanding one's role and strategy within a larger ecosystem.

Video

- **The Li & Fung Digital Platform (02:09)**



3.3

What is the impact of information systems on organizations?

Economic Impacts

- **Information Technology's Role in Economics**
  - IT changes both the relative costs of capital and the costs of information
  - Acts as a substitutable production factor for traditional capital and labor

- **Substitution for *Labor***
  - Decreasing IT costs lead to ***labor substitution***.
  - Predicted decline in middle management and clerical roles.

- **Substitution for *Capital***
  - IT also replaces more expensive forms of capital (e.g., buildings, machinery).
  - Managers are likely to boost IT investments as its cost effectiveness increases

Economic Impacts: Transaction Cost Theory

- **Transaction Cost Theory**
  - Emphasizes ***minimizing*** transaction costs alongside production costs.
  - Includes search and information costs, bargaining and decision costs, policing and enforcement costs.
  - Traditional firms reduced costs through ***vertical integration***, hiring more employees, buying suppliers and distributors.
- **Networks and digital connections *lower* market transaction costs while encouraging outsourcing over vertical integration**

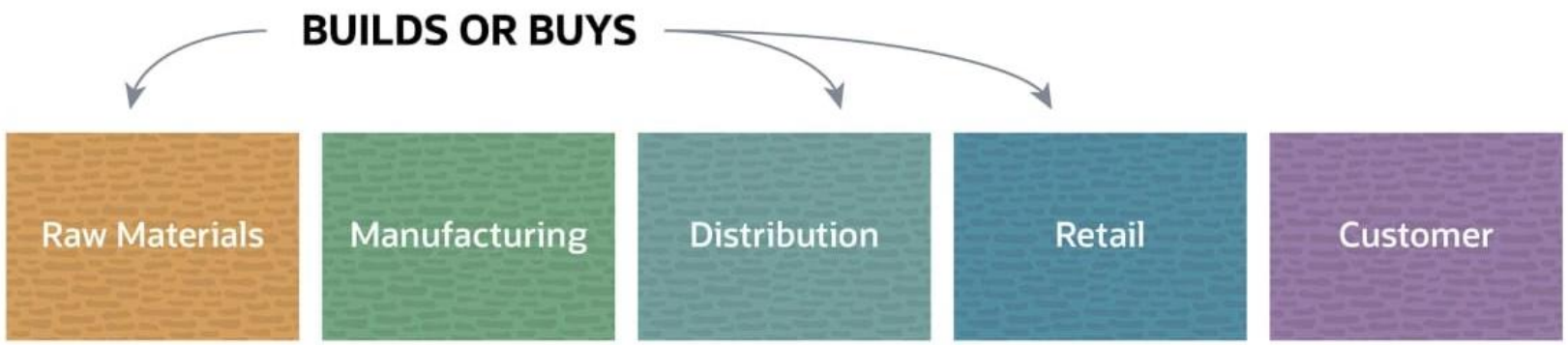


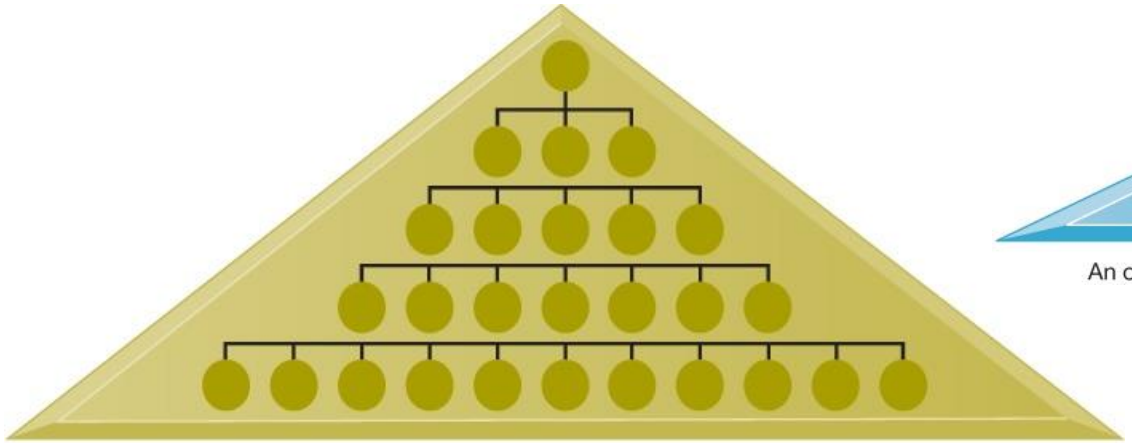
Image Source: NetSuite

Economic Impacts: Agency Theory

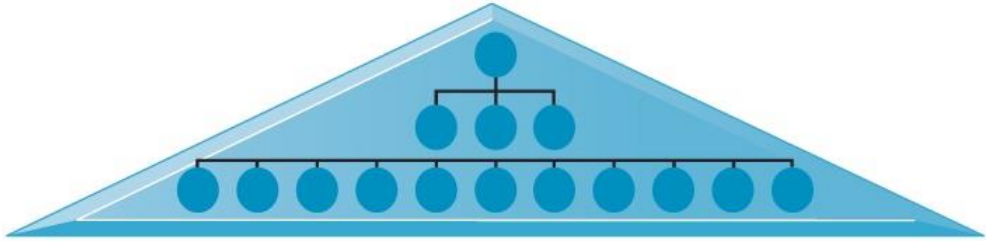
- **Agency Theory (Jensen and Meckling, 1976)**
  - Firms as a nexus of contracts among individuals
  - Growth increases the need for supervision, raising management costs
- **Agents (employees) require constant supervision to *align* their interests with those of the owners.**
  - Agency costs rise with firm size and complexity.
- **IT *reduces costs of supervision* by making information more accessible, and enables managers to oversee efficiently.**
- **Impact on Organizational Structure**
  - _____in the number of middle managers and clerical staff
  - Potential for firm size to _____ while increasing revenue per employee

IT flattens organizations

- **IT facilitates *flattening* of hierarchies by broadening the distribution of information to empower lower-level employees and increase management efficiency**
  - Decision making is pushed to ***lower levels***
  - Fewer managers are needed
  - Management costs decline
  - Hierarchy becomes much more efficient



A traditional hierarchical organization with many levels of management



An organization that has been "flattened" by removing layers of management

Organizational Impact and Resistance

- **Organizational Impact**
  - Leads to ***simpler*** business processes and ***flatter*** organizational structures
    - Firms are ***restructuring key business processes*** around Internet technology ➔ ?
    - Integration into IT infrastructures for streamlined operations
  - Potential ***reduction*** in employee numbers due to efficiency gains

- **Organizational and political resistance often hinders IT project success**

- **Managerial ability to work with people is as crucial as technical knowledge**

3.4

What are the challenges posed by strategic information systems, and how should they be addressed?

Sustaining Competitive Advantage

- **Strategic information systems can reshape business processes and behaviors, but success requires *alignment* among technology, organizational structure, and management**

- **Competitive advantage from strategic systems is often *temporary*, as rapid technological change, globalization, and the Internet make imitation easier**

- **Information systems alone do not ensure lasting advantage; sustained success depends on *continuous innovation and organizational adaptation***

Aligning IT with business objectives

- **Research on IT and business performance has found that**
  - (a) the more successfully a firm can ***align*** information technology with its business goals, the more profitable it will be
  - (b) only one-quarter of firms achieve ***alignment*** of IT with the business
  - (c) about half of a business firm's profits can be explained by the alignment of IT with business

- **Successful firms and managers understand what IT can do and how it works, take an active role in shaping its use, and measure its impact on revenues and profits**

Key Takeaways

- Managers must understand organizational structures, processes, culture, and incentives to successfully build and use IS.
- Information systems can significantly reshape organizational processes, decision-making, and performance, often driving organizational change.
- Strategic frameworks such as Porter's competitive forces, the value chain, core competencies, synergies, and network economics help firms use information systems to gain competitive advantage.
- Strategic information systems pose challenges such as imitation, rapid technological change, and misalignment, which require continuous adaptation and strong managerial oversight.

Next Week...

- **Next week, we will cover:**
  - [Chapter 9] Achieving operational excellence and customer intimacy

Q & A